DIGITAL LOGIC PROJECT

Digital Door Access System Using Combination Logic



Presented by Team - 07

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INTRODUCTION

Introduction and objective

The problem

Conventional keys can be lost, copied, or shared without visibility.

The idea

A digital door access system takes an entered code and decides, using logic gates, whether to unlock.

The approach

Build the decision circuit from basic gates, simulate it in Logisim and verify it with a truth table.



Objective: design and simulate a combinational logic circuit that unlocks the door only for the valid input code (A = 1, B = 0, C = 1) and rejects every other combination.

Access control

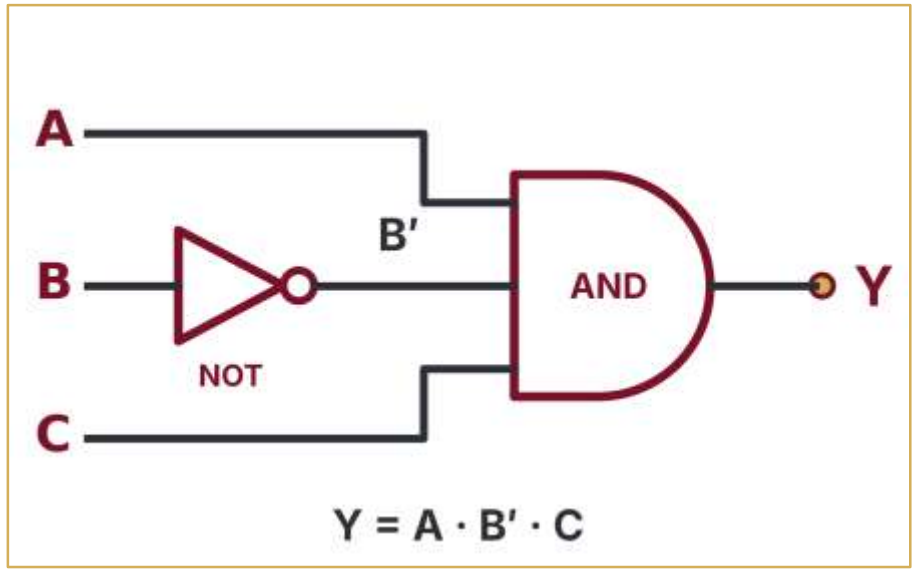
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THEORY

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LOGIC GATES

Theory and logic circuit



1. AND gate: the output is 1 only when all of its inputs are 1.

2. NOT gate: inverts its input, so 0 becomes 1 and 1 becomes 0.

3. B goes through a NOT gate, then A, NOT B and C enter a 3-input AND gate.

4. Y = A · B′ · C, so the door unlocks only for A = 1, B = 0, C = 1.

Combinational logic: the output depends only on the present inputs, with no memory.

Gates & Boolean logic

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OPERATION

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SYSTEM FLOW

How the access system works



Valid code entered

A = 1, B = 0, C = 1: the NOT gate makes B′ = 1, so all three AND inputs are 1.

Y = 1 and the door unlocks.

Any other code

At least one AND input is 0, so Y = 0 and the door stays locked.

The A = B = C = 0 test in Logisim is one such case.

Combinational design: the decision follows the inputs instantly and stores no state.

Input to output

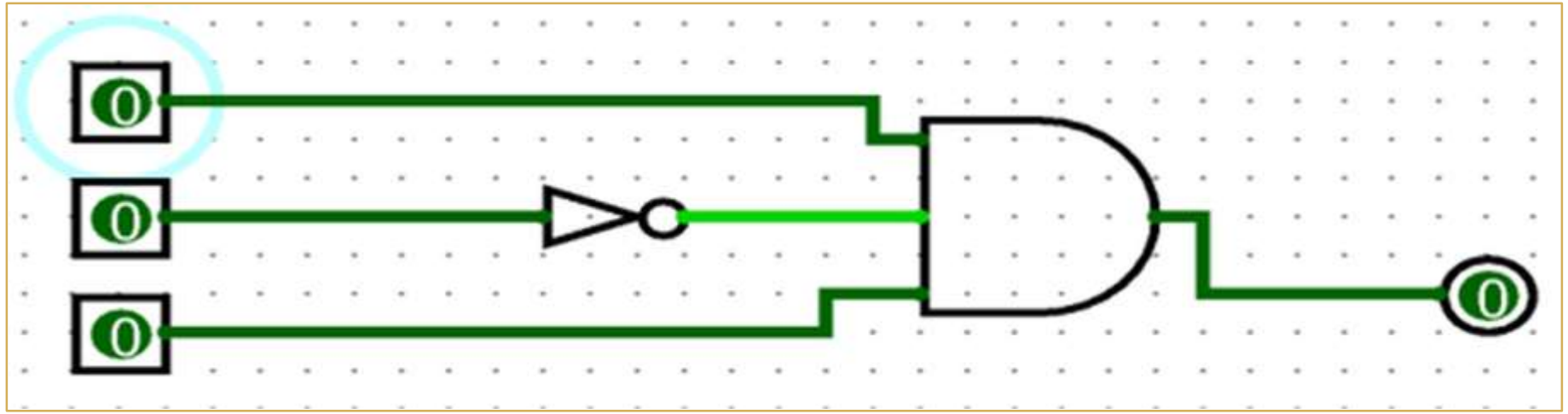
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LOGISIM

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CIRCUIT DESIGN

Circuit design in Logisim



Components used

Three input pins (A, B, C), a NOT gate on B, a 3-input AND gate and one output pin Y.

Bright green wires carry logic 1; dark green wires carry logic 0.

Working of the circuit

B is inverted, so the AND gate receives A, B′ and C.

Y = 1 only when A = 1, B = 0 and C = 1; that output unlocks the door.

Logisim shows every wire’s logic level live, so the design can be tested one input pattern at a time.

Built in Logisim

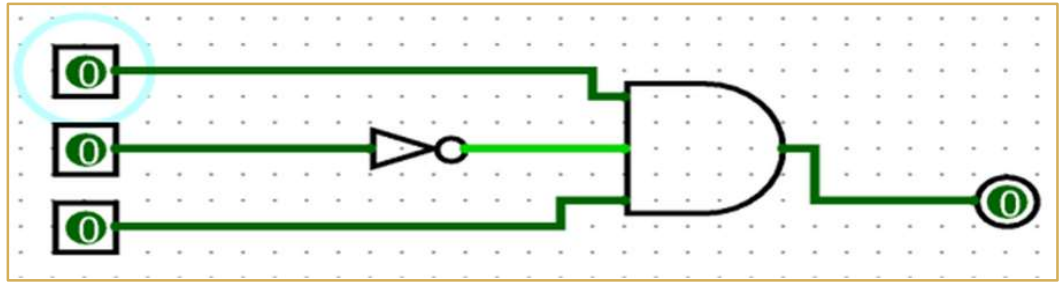
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SIMULATION

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RESULTS

Simulation results



Logisim run: A = B = C = 0 gives Y = 0 (access denied).

What the simulation shows

With A = B = C = 0, the NOT gate makes B′ = 1, but A and C are 0, so Y = 0.

Of the 8 input combinations, only A = 1, B = 0, C = 1 gives Y = 1.

Result: the simulation matches the truth table; only code 101 unlocks the door.

A	B	C	B′	Y
0	0	0	1	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	0
1	0	0	1	0
1	0	1	1	1
1	1	0	0	0
1	1	1	0	0

Case shown in the Logisim screenshot

The only valid code (Y = 1)

Truth table

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CONCLUSION

Outcome and scope

Conclusion and applications

01

Outcome

A 3-input access circuit was designed and simulated in Logisim.

Output Y = 1 only for A = 1, B = 0, C = 1.

The other seven input combinations are denied.

02

Applications

Home and office door locks

Keypad and digital-code entry panels

Lockers, safes and restricted rooms

03

Improvements

Extend to a longer 4-bit or 8-bit access code.

Add flip-flops to count wrong attempts and raise an alarm.

Drive a real lock through a microcontroller or FPGA.

Result: a simple, testable combinational design that shows how basic gates make a door-access decision.

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Thank You



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GitHub link - https://github.com/Neelagreeva/DDCA-Digital-Door-Access-System-by-Team---07